FlowMind AI

An Evidence-Grounded, Approval-Gated Agent for a Single High-Stakes Enterprise Workflow

Tagline: Think. Retrieve. Reason. Recommend. Approve. Act. Track.

Final-Year Project Proposal

Core Concept

FlowMind AI is a focused enterprise workflow intelligence prototype that converts a natural-language request into an evidence-grounded, auditable action cycle. Instead of functioning as a generic document chatbot, the system retrieves relevant organizational evidence, reasons over that evidence, recommends a next-best action with justification, obtains human approval for sensitive operations, executes an approved workflow, and records the outcome for traceability. The project is built around a single, fully implemented workflow and a specific, testable research question, rather than a broad multi-workflow platform.

Primary workflow: Request → Retrieve Evidence → Reason → Recommend → Human Approval → Execute → Track Outcome → Audit

1. Research Question & Core Hypothesis

This project is framed as a focused case study that tests a specific, measurable claim, rather than simply positioning FlowMind AI as a general enterprise automation platform.

Research question: Does closing the loop — evidence retrieval → grounded recommendation → human approval → controlled action → audit — improve task success rate and user trust, compared to a standard RAG chatbot that only retrieves and answers?

Hypothesis: An evidence-grounded, approval-gated workflow system (FlowMind) leads to a higher task success rate and greater user trust than a plain RAG chatbot that only retrieves and answers questions.

2. Abstract

Organizations store operational knowledge across documents, support tickets, emails, databases, and business applications. Employees often spend significant time searching for context, deciding what to do next, switching between systems, and manually performing repetitive actions. FlowMind AI addresses this problem through a unified agentic workflow intelligence system. The proposed prototype combines Retrieval-Augmented Generation (RAG), controlled AI reasoning, evidence-backed next-best-action recommendations, human-in-the-loop approval, workflow execution, and audit logging.

The project will deliberately focus on a limited enterprise domain and implement one complete, high-quality end-to-end workflow (Customer Complaint Escalation) rather than attempting to support every department or application. Two further workflows are retained in the proposal as an argument for generalizability, but are not built. The system will provide citations to retrieved evidence, explain the basis of recommendations, restrict actions through role-based access control, and record each important decision and execution step. Its effectiveness will be evaluated using retrieval quality, task success rate, execution accuracy, response time, audit completeness, and a small human evaluation comparing FlowMind against a plain-RAG baseline.

3. Problem Statement

Enterprise search, workflow automation, and decision support are frequently handled by separate tools. A user may need to search multiple sources, interpret fragmented information, decide on an action, open another application, perform the action, and manually document the result. Basic enterprise chatbots can retrieve or summarize information but often stop before controlled action execution. Traditional automation tools can execute predefined workflows but may lack contextual understanding of unstructured information.

There is therefore a need for a focused system that can connect evidence retrieval, contextual reasoning, explainable recommendations, controlled human approval, action execution, outcome tracking, and auditing within one workflow.

4. Scope and Positioning

The project will be implemented as a proof-of-concept for one enterprise domain, customer support and operations. It will demonstrate one fully built, end-to-end workflow using a controlled set of documents, tickets, and a simulated enterprise connector. Two additional workflows are described in Section 9 as evidence that the architecture generalizes, but they are not implemented. The goal is not to claim a production-ready platform for all departments. The goal is to demonstrate a technically complete architecture and test whether evidence-grounded, approval-gated AI measurably outperforms a plain RAG chatbot at bridging the gap between finding information and taking an authorized action.

5. Key Differentiator

The main contribution is not merely combining a chatbot with RAG. FlowMind AI closes the workflow loop. Every recommendation must be linked to retrieved evidence; important actions require authorization or human approval; executed actions are tracked; and the full chain is stored in an audit log. A standard RAG chatbot, by contrast, typically stops at retrieving and summarizing information: it does not generate an actionable, evidence-linked recommendation, does not gate actions behind approval, and does not execute or audit anything. The system therefore demonstrates an observable sequence from organizational knowledge to decision to measurable action, and this project measures whether that sequence is actually better than the plain-retrieval alternative, not just different from it.

6. Objectives

- Develop a conversational interface that accepts natural-language enterprise requests.
- Implement a RAG pipeline that retrieves relevant evidence from a controlled enterprise knowledge base and provides source references.
- Build an orchestration layer that classifies intent and selects retrieval, recommendation, approval, and action tools.
- Generate evidence-backed next-best-action recommendations rather than making unsupported claims of mathematically optimal decisions.
- Provide concise explanations showing why a recommendation was generated and which evidence supports it.
- Implement human-in-the-loop approval before sensitive or high-impact actions are executed.
- Execute the approved workflow through a mock connector / simulated enterprise API.
- Implement authentication, role-based access control, action permissions, and audit logging.
- Track workflow outcomes and evaluate the prototype using measurable technical and operational metrics.
- Compare FlowMind against a plain-RAG chatbot baseline on retrieval quality, task success rate, and user-rated trust.

7. Proposed End-to-End Workflow

- **1. Request:** User submits a natural-language task or problem.
- **2. Intent & Planning:** Orchestrator identifies the goal and determines required retrieval and tools.
- **3. Evidence Retrieval:** RAG retrieves relevant documents, tickets, policies, or records.
- **4. Evidence-Grounded Reasoning:** The reasoning layer analyzes only the permitted and retrieved context.
- **5. Next-Best-Action Recommendation:** The system proposes an action with evidence, rationale, and an optional confidence indicator.
- **6. Human Approval:** Sensitive actions are displayed for approval, rejection, or modification.

- **7. Controlled Execution:** The automation engine executes only approved and authorized actions.
- **8. Outcome Tracking:** Execution status and resulting changes are captured.
- **9. Audit:** Request, evidence references, recommendation, approver, action, timestamps, and outcome are logged.

8. Demonstration Workflow (Primary, Fully Built)

Customer Complaint Escalation

User asks the system to investigate a repeated customer complaint. FlowMind retrieves related tickets and policies, identifies unresolved issues and the responsible team, recommends an escalation with justification, asks for approval, creates the escalation task through a mock connector, and logs the outcome. This workflow is chosen as the single fully-built demonstration because it has the clearest evidence trail, the simplest action surface, and the most intuitive approval decision for a human evaluator to judge.

9. Generalizability (Argued, Not Built)

The following two workflows are retained in the proposal as evidence that the architecture is not workflow-specific. Both would reuse the same five modules — retrieval, evidence-grounded reasoning, recommendation, approval, execution — with only the connector and prompt templates changing. Neither is implemented in the final-year build.

SLA Risk and Support Prioritization: FlowMind retrieves ticket history, priority rules, and SLA policy, identifies cases at risk of breach, recommends which case should be handled next and why, obtains approval where required, updates priority or assigns the ticket, and records the action.

Internal Operational Request: An employee submits a request such as access, procurement, or administrative support. FlowMind retrieves the applicable policy and previous request context, checks required conditions, recommends the next step, routes the request for approval, executes an allowed action or creates the appropriate task, and tracks completion.

10. System Modules

- **Conversational Interface:** User requests, session context, evidence display, approval UI, and execution results.
- **Reasoning & Orchestration:** Intent classification, workflow planning, tool selection, state management, and guardrails.
- **Retrieval Intelligence:** Document ingestion, chunking, embeddings, vector retrieval, metadata filtering, reranking, and source citation.
- **Decision Support:** Produces evidence-backed next-best-action recommendations with rationale; it does not claim guaranteed optimality unless an explicit optimization method is implemented.
- **Human-in-the-Loop Approval:** Prevents sensitive actions from being automatically executed without an authorized decision.
- **Automation Engine:** Executes approved tasks through a mock connector / simulated enterprise API.
- **Security & Access:** Authentication, RBAC, tool permissions, and data-access restrictions.
- **Audit & Outcome Tracking:** Stores evidence references, recommendations, approvals, execution events, errors, and final outcomes.

11. Suggested Architecture

Layer	Suggested Implementation
Frontend	React / TypeScript

Backend & APIs	FastAPI (Python)
AI Orchestration	Python orchestration layer; optional agent/RAG framework
LLM	GPT-class or Claude-class API, with a replaceable provider interface
Embeddings / Retrieval	Embedding model + vector search; metadata filtering and optional reranking
Operational Database	PostgreSQL
Vector Storage	pgvector or a dedicated vector database
Authentication & Security	JWT/session authentication + RBAC + action permissions
Automation	Mock / simulated enterprise connector (single workflow)
Deployment	Docker; local or cloud prototype
Observability	Structured logs, workflow status, audit events, latency/error tracking

12. Evaluation and Measurable Outcomes

- **Retrieval Quality:** Precision@K, Recall@K, or manually labelled relevance for a test question set.
- **Grounded Answer Quality:** Percentage of responses correctly supported by retrieved evidence.
- **Workflow Task Success Rate:** Percentage of test workflows completed with the expected final state.
- **Action Execution Accuracy:** Percentage of actions executed with correct parameters and target system.
- **Human Approval Compliance:** Verify that protected actions cannot bypass the approval step.
- **Response / Workflow Latency:** Measure retrieval, recommendation, and end-to-end execution time.
- **Audit Completeness:** Percentage of workflow runs containing all required evidence, decision, approval, execution, and outcome records.

Human evaluation (comparison against a plain-RAG baseline)

5–10 participants review a sample of FlowMind's and a plain-RAG baseline's outputs, blind and unlabelled, and rate each on trust, clarity of justification, and perceived usefulness on a short Likert-scale survey. This adds a human-centred dimension to the otherwise systems-metrics evaluation and directly supports the hypothesis in Section 1.

13. Example Evaluation Scenario

A benchmark dataset can contain sample company policies, 100–500 synthetic or anonymized support tickets, and predefined workflow cases with expected outcomes. For each test case, the system is evaluated on whether it retrieves the correct supporting evidence, recommends an appropriate action, requests approval when required, invokes the correct tool with valid parameters, and records the final outcome. The same test cases are run through the plain-RAG baseline for direct comparison. This gives the project measurable results beyond a visual chatbot demonstration.

14. Security and Responsible AI Design

- **Role-based retrieval:** users can retrieve only information permitted for their role.

- **Action-level authorization:** access to information does not automatically grant permission to execute an action.
- Human approval for sensitive, irreversible, or high-impact actions.
- Evidence citations so users can verify the context behind recommendations.
- Audit logs containing actor, timestamp, recommendation, approval decision, tool invocation, and result.
- Graceful abstention when evidence is missing, contradictory, or insufficient.
- Protection against prompt instructions contained inside retrieved documents being treated as trusted system commands.

15. Phased Build Plan (16 Weeks)

Phase	Weeks	Focus	Checkpoint
1	1–5	Data & Retrieval Foundation: prepare 100–200 synthetic tickets and policy documents; build ingestion, chunking, embeddings, and vector retrieval with source citations.	Retrieval demo with Precision@K on a labelled test set
2	6–9	Recommendation Layer + Baseline: build the evidence-linked recommendation module with rationale generation; build the plain-RAG baseline in parallel.	Recommendation output reviewed against 10–15 hand-labelled cases
3	10–13	Approval, Execution & Audit: build the approval UI, mock execution connector, RBAC, and audit log.	Full closed-loop demo, one case end to end
4	14–16	Evaluation & Write-Up: run test cases through FlowMind and the baseline, collect technical metrics, run the human evaluation, finalize the report.	Final metrics table and evaluation chapter draft

16. Expected Outcomes

- A functional prototype demonstrating one complete, evidence-grounded, approval-gated enterprise workflow.
- A plain-RAG baseline built for direct comparison.
- Evidence-grounded answers and recommendations with traceable sources.
- Controlled automation with human approval and role-based permissions.
- Measurable evaluation of retrieval, recommendation, execution, latency, and audit completeness, alongside a small human evaluation of trust and clarity.
- A written argument, supported by Section 9, for how the architecture would extend to additional enterprise workflows.

17. Explicit Non-Goals for the Final-Year Prototype

The prototype will not attempt to replace enterprise-wide RPA, BI, search, or decision systems; integrate dozens of third-party products; guarantee globally optimal business decisions; autonomously execute every action without oversight; or build and evaluate all three workflows described in this proposal — only one is fully built, and two are argued for generalizability. These boundaries keep the project technically achievable while preserving meaningful depth.

18. Future Scope

- Add production-grade connectors for enterprise applications and data sources.
- Introduce predictive models where sufficient historical outcome data is available.
- Use outcome feedback to improve recommendation ranking.
- Explore multi-agent coordination for complex cross-department workflows.
- Add on-premise or private-cloud deployment for data-sensitive organizations.
- Introduce formal optimization algorithms for specific decision problems where an objectively defined optimal solution is possible.
- Build out the SLA Risk Prioritization and Internal Operational Request workflows described in Section 9.

19. Related Work / Positioning

FlowMind is positioned against existing enterprise AI assistants such as Moveworks, Glean, and ServiceNow Virtual Agent, which provide retrieval and, in some cases, action execution. The contribution of this project is not a new product category, but a transparent, auditable approval gate placed between recommendation and execution, and a measured comparison of grounded, approval-gated recommendation against a plain RAG baseline — evaluation data of a kind that these commercial tools do not disclose publicly.

20. Final Project Definition

FlowMind AI is an agentic enterprise workflow intelligence prototype that transforms a natural-language request into an evidence-grounded and auditable action cycle. It retrieves authorized organizational knowledge, generates an explainable next-best-action recommendation, obtains human approval for protected operations, executes an authorized workflow, and tracks measurable outcomes.

This refined definition gives the project a clearer technical identity than a generic RAG chatbot, and a testable research question: does closing the loop from evidence to recommendation to approved action to outcome improve task success and user trust compared to a plain RAG chatbot? The final demonstration should emphasize the complete closed-loop workflow and support its claims with quantitative evaluation and human feedback.

. References

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